

Summer Internship Project: Clinical Decision Support — Early Warning & Medication Safety

Project Description

Every day in hospital wards, patients deteriorate quietly before anyone notices. Sepsis, acute kidney injury, and impending cardiac arrest each leave a trail of subtle signals in vitals and lab trends — signals that nurses and residents rarely have time to synthesise manually across a full ward of patients. At the same time, drug-lab interactions slip through unchecked when a new prescription is written without cross-referencing the most recent potassium level or creatinine trend.

This project builds an AI-powered ward monitoring console designed for nurses — the clinicians who are closest to the data and best positioned to act early. You will develop a real-time dashboard that auto-calculates early warning scores, runs a machine learning deterioration model over time-series vitals and labs, and continuously checks each patient's active medications against their current lab panel for dangerous interactions.

You will train and evaluate models on MIMIC-IV and the PhysioNet Sepsis Challenge dataset, build on validated open-source pipelines, and deliver a live demo that replays real ICU stays through the system. The goal is a prototype compelling enough for hospital nursing and clinical informatics teams to share real ward data for validation.

Duration: 8–10 weeks | **Team size:** 2 students | **Domain:** Clinical Decision Support & Patient Safety

Target Users

User	Role in Prototype
Bedside nurses	Primary users — monitor patient acuity list, act on deterioration and drug-lab alerts
Charge nurses	Triage escalations, manage ward-level risk overview
Attending physicians / residents	Secondary recipients of escalation alerts generated by the system
Clinical informatics teams	Evaluate alert sensitivity/specificity before hospital adoption

Problem Statement

- Manual and intermittent early warning score calculation means deteriorating patients are frequently identified too late
- Sepsis and AKI have 4–6 hour windows where early intervention meaningfully changes outcomes — current systems miss these
- Drug-lab interactions (e.g., ACE inhibitors with rising K⁺, metformin with falling eGFR) rely entirely on physician recall at prescribing time

- Nurses administer medications and chart vitals but have no decision support beyond paper escalation protocols
 - In-hospital cardiac arrests and sepsis-related mortality remain significantly higher in Indian hospitals than international benchmarks
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Prototype: Real-Time Ward Monitoring and Alert Console

The student team will deliver a **live web dashboard** designed to sit on the nurses' station screen, updated continuously as new vitals and lab values are charted. The system ingests a simulated real-time vitals feed (replaying MIMIC-IV ICU time-series at accelerated speed) so the demo feels live rather than static.

Module A — Early Warning Score Engine & Deterioration Predictor

What it does: The central panel of the dashboard is a **ranked patient list** for the ward, ordered by current deterioration risk. Each patient row shows:

- Auto-calculated **NEWS2 / MEWS score** from the latest charted vitals
- A **6-hour trend sparkline** for key vitals (HR, RR, SpO₂, BP, Temperature, Consciousness)
- A **colour-coded ML risk flag** (green / amber / red) generated by a gradient boosting or LSTM model trained to predict clinical deterioration 4–6 hours before the event

The ML model uses *combinations* of trending values — not single threshold breaches — catching patients who are sliding gradually rather than those who have already crossed a panic value. When a patient crosses amber, the system generates a structured **escalation prompt**:

- Patient name and bed number
- The specific risk factors driving the score (e.g., "RR trending from 16 to 22 over 4 hours; SpO₂ drifting; new low-grade fever")
- Recommended nursing response (e.g., "Repeat full obs set in 15 minutes; notify on-call resident if score unchanged")

A **secondary panel** shows patients meeting **Sepsis-3** or **KDIGO AKI** criteria, with the time of first criterion breach displayed so response latency can be measured retrospectively — a metric hospitals can use to demonstrate improvement post-deployment.

What the student builds:

- NEWS2/MEWS score calculator from raw vitals (rule-based, fully transparent)
 - LSTM or XGBoost deterioration predictor trained on MIMIC-IV time-series
 - Sepsis-3 and AKI-KDIGO labelling pipeline (using validated MIT-LCP SQL extractions)
 - Real-time patient acuity dashboard with sparklines and escalation prompts
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Module B — Drug–Lab Interaction Checker

What it does: A background process runs continuously against each patient's **active medication list** and their **most recent lab panel**. It applies a rule library — sourced from RxNorm and the FDA drug interaction database — to surface interactions requiring nurse awareness before the next dose is administered.

Example alerts generated:

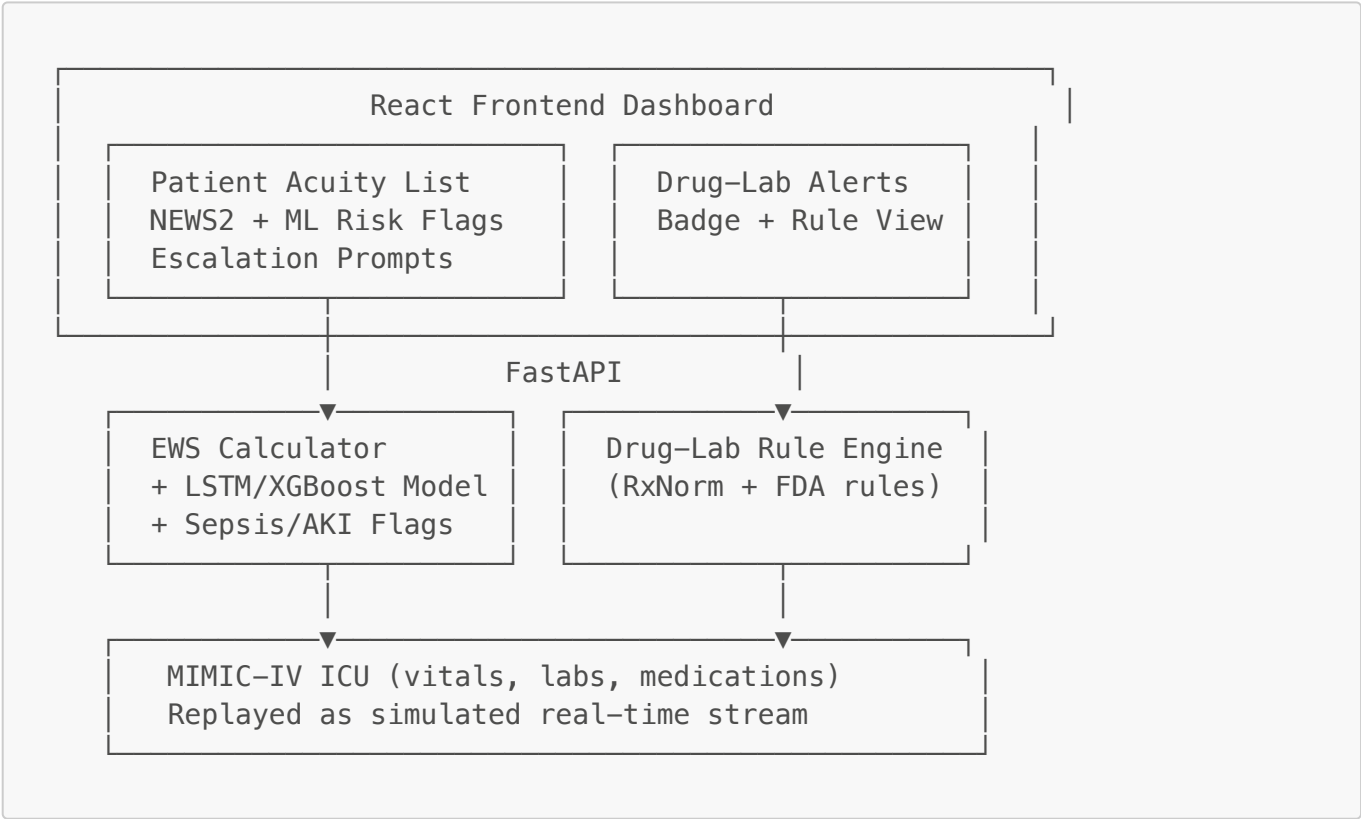
Trigger condition	Alert shown
K ⁺ > 5.5 mEq/L + active ACE inhibitor or K-sparing diuretic	⚠ Hyperkalaemia risk — hold next dose, notify prescriber
eGFR < 30 + active metformin	⚠ Lactic acidosis risk — dose review required
Creatinine trending upward + active NSAID	⚠ AKI risk — flag for prescriber review
INR > 3.5 + active anticoagulant	⚠ Bleeding risk — check dosing schedule

Alerts appear as a **badge on the patient row** with a one-line plain-English explanation and the specific lab value that triggered it. Every alert shows its **full rule logic** — this transparency is intentional, making the system auditable and accelerating nurse and physician trust compared to black-box scoring.

What the student builds:

- Structured rule library (JSON/YAML) sourced from RxNorm + FDA interaction database
- Rule engine that cross-references active med list against latest labs per patient
- Alert badge UI integrated into the ward dashboard, with rule explanation on click

Technical Architecture



Datasets & Starting Points

Resource	Use
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Resource	Use
MIMIC-IV icu module — PhysioNet	Hourly vitals, labs, medication records for ICU cohorts
MIT-LCP/mimic-code	Validated SQL/Python extractions for Sepsis-3, AKI-KDIGO, NEWS scores
YinchangChang/DeepEWS	LSTM-based early warning score baseline model
MLforHealth/MIMIC-Extract	Preprocessed time-series pipeline ready for model training
PhysioNet Sepsis Challenge 2019	Labelled sepsis onset timestamps; purpose-built for early prediction
RxNorm API — NLM	Drug normalisation and interaction rule sourcing
FDA Drug Interaction Database	Rule sourcing for the drug-lab checker
clinicalml/clinical-deterioration	Deterioration prediction framework from MIT Clinical ML Group

Evaluation Metrics

Module A — Deterioration Predictor

- AUROC and AUPRC on MIMIC-IV held-out test set (target: AUROC > 0.80)
- Alert lead time: median hours between first amber flag and recorded clinical event
- False positive rate at fixed sensitivity thresholds (nurse workload impact)
- NEWS2 score accuracy vs. manually computed ground truth

Module B — Drug-Lab Checker

- Rule coverage: % of known clinically significant interactions from reference list captured
- Alert precision: % of alerts triggered that are clinically valid on chart review
- Alert timeliness: % of alerts generated before the implicated dose was administered (on MIMIC retrospective replay)

Demonstration Plan for Hospital Partners

Replay 50–100 historical MIMIC ICU stays through the live dashboard during the demo, showing the system flagging deterioration events in real time — hours before the recorded clinical response in the notes. Overlay the drug-lab alerts on the same stays and show how many would have been surfaced before the dose was administered.

The ask at the end of the demo:

"Give us access to 3 months of de-identified vitals, lab, and medication administration records from two general wards so we can validate alert sensitivity and specificity on your patient population before any go-live discussion."

Deliverables

- ☐ Working real-time dashboard prototype hosted locally via Docker
- ☐ Trained deterioration model with evaluation report on MIMIC test set
- ☐ Drug-lab rule library (JSON/YAML) with source citations for each rule
- ☐ Alert audit log (per-patient, per-event) for retrospective review
- ☐ 10-minute recorded demo video of a live MIMIC ICU replay
- ☐ Clean GitHub repository with setup instructions
- ☐ 2-page project summary document for hospital nursing and informatics teams